RWTH AACHEN
UNIVERSITY

36

Overview

Introduction of retinal prosthetics

Encoding strategies

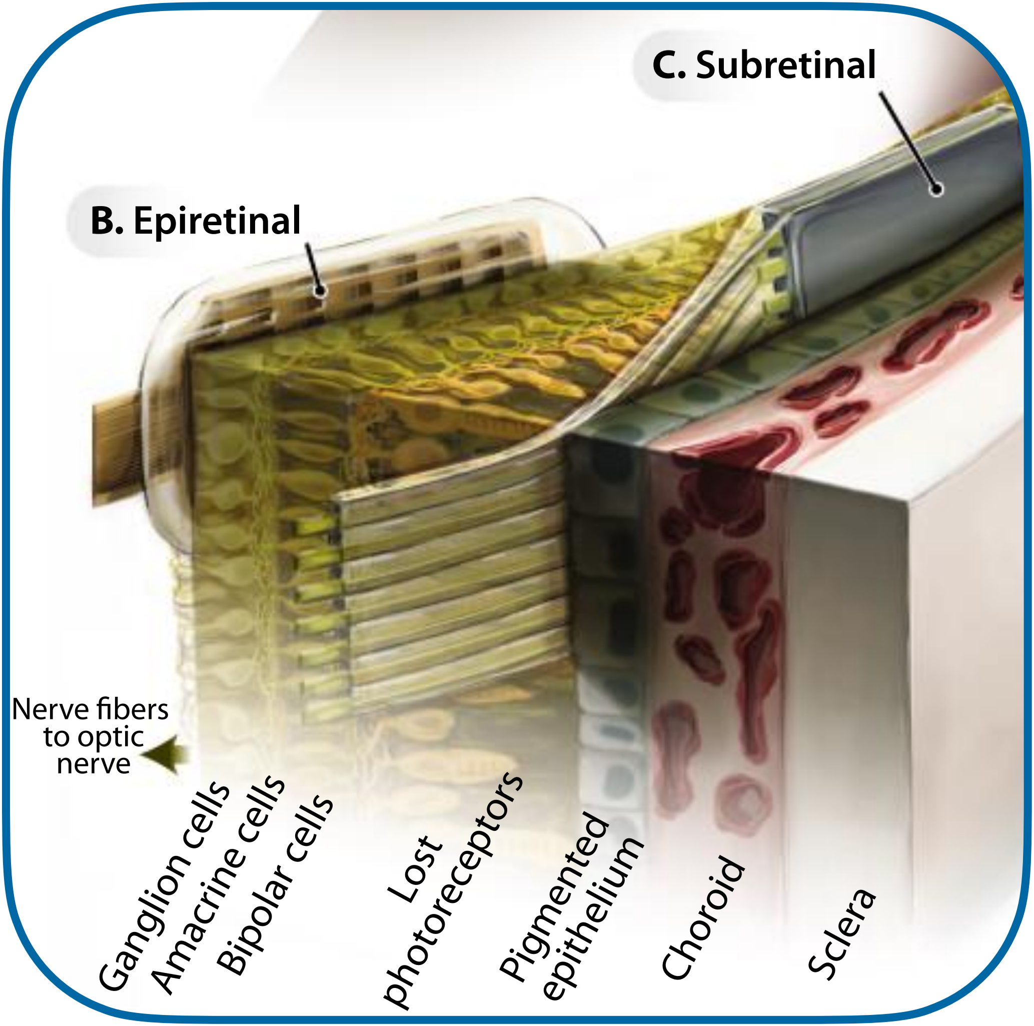
- Reverse encoding
- End-to-end encoding
- Invertible encoding

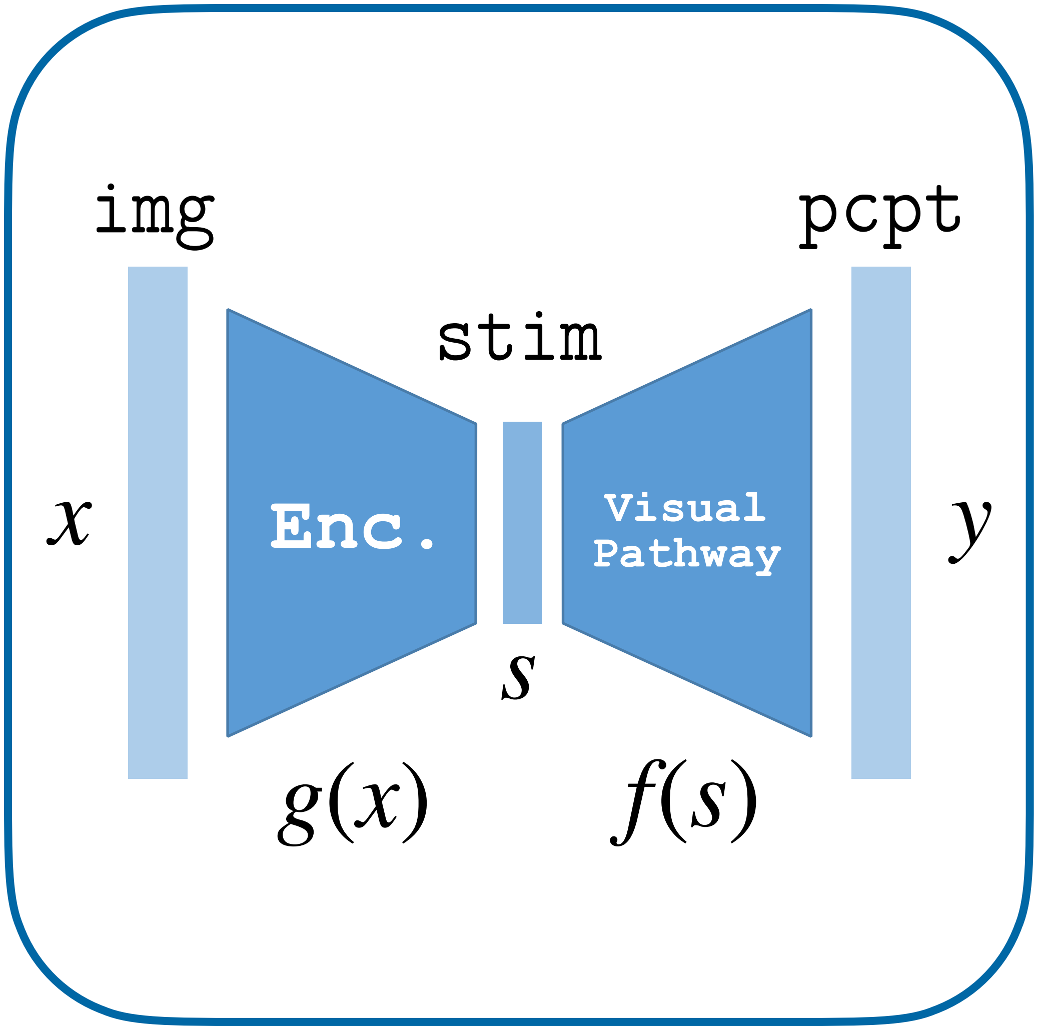
Eye gaze

- Visual fixation-based simulation
- Visual fixation consistency

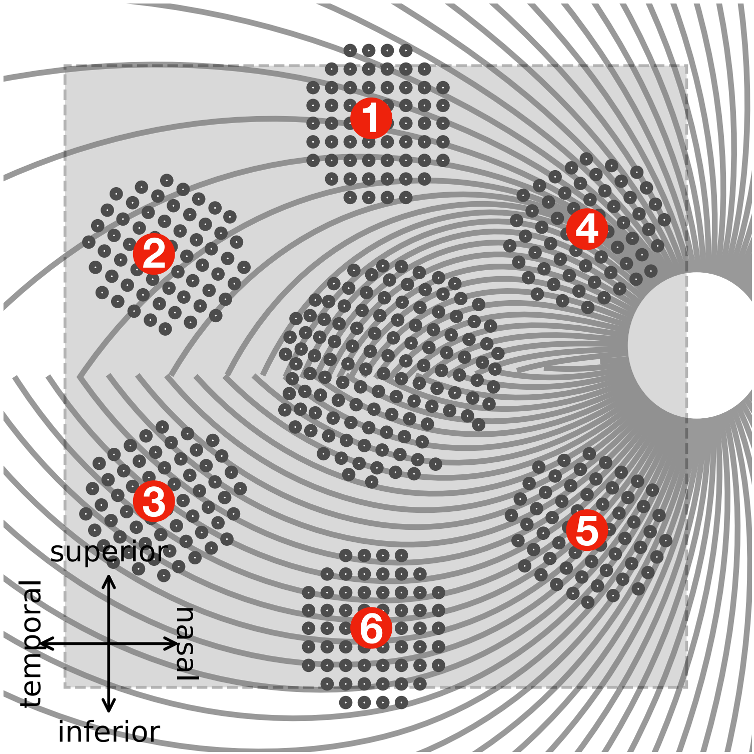
Supplementary

- Questionnaire
- Braille simulation


C. Subretinal
B. Epiretinal
Nerve fibers
to optic
nerve
Ganglion cells
Amacrine cells
Bipolar cells
Lost
photoreceptors
Pigmented
epithelium
Choroid
Sclera


img
pcpt
stim
x
Enc.
Visual
Pathway
y
s
g(x)
f(s)


0.00s



1
2
3
4
5
6
superior
temporal
nasal
inferior


0.00s


0.00s

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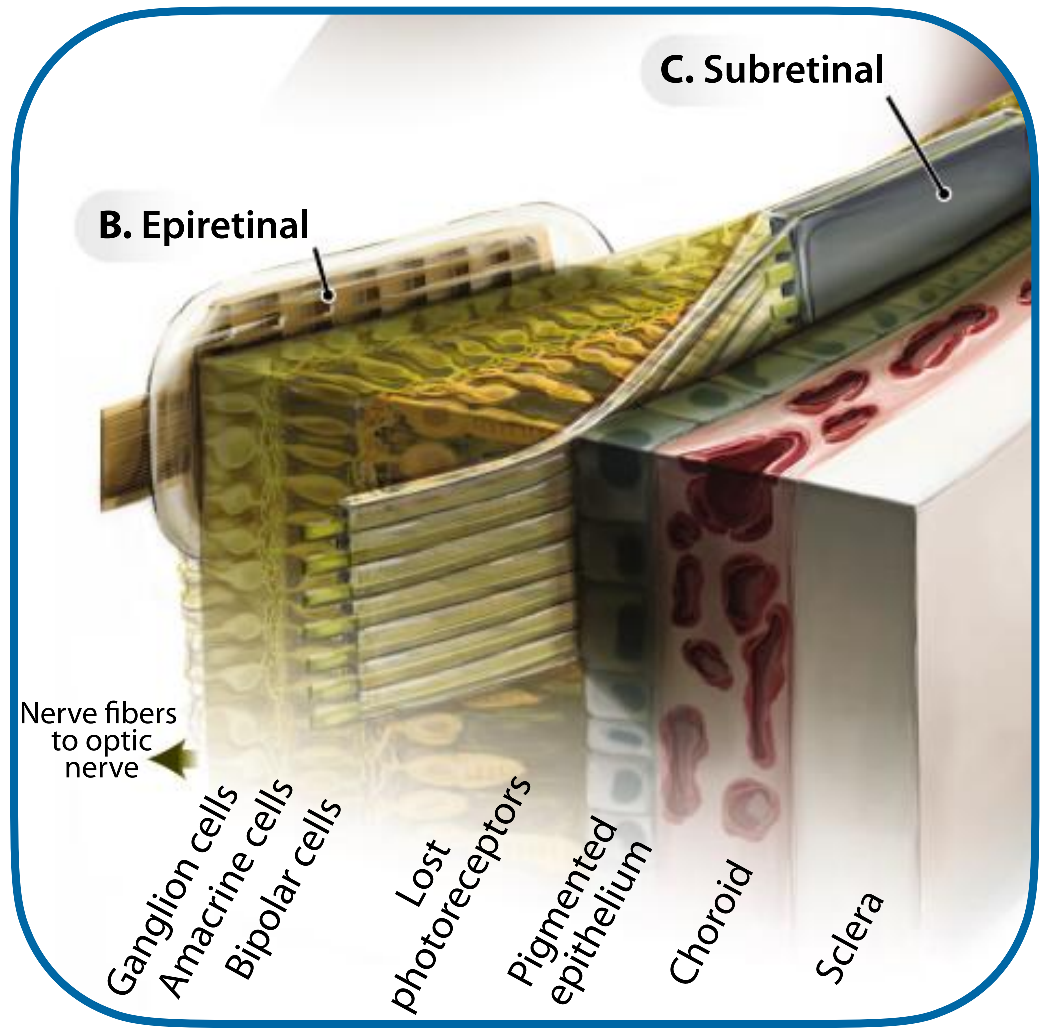
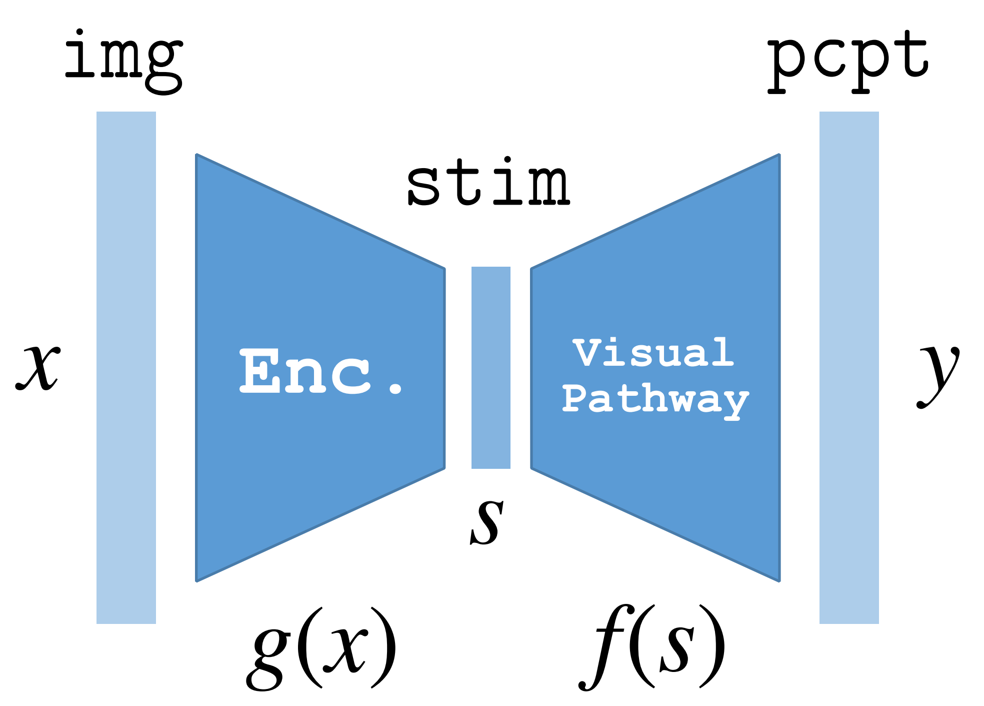
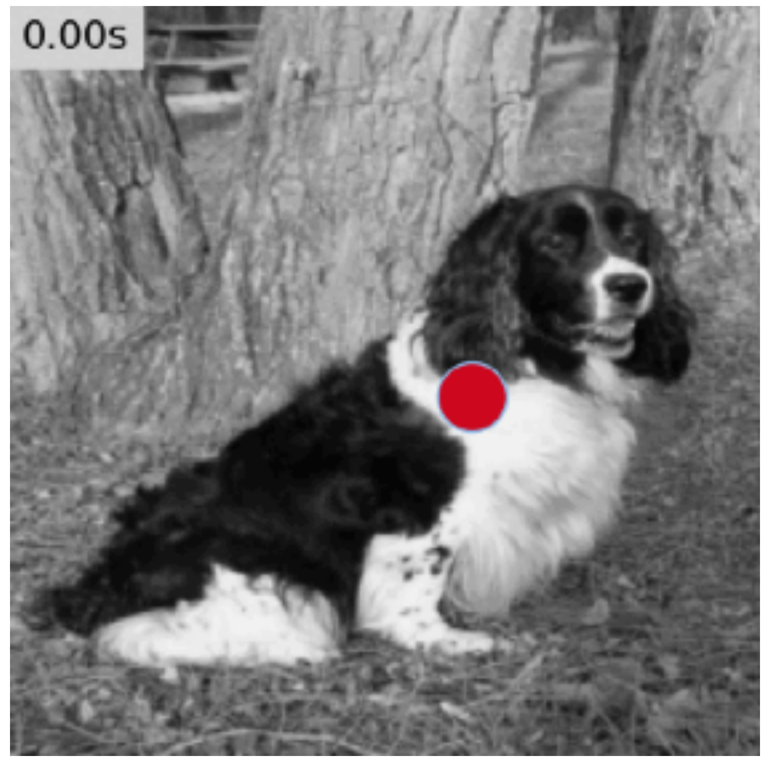
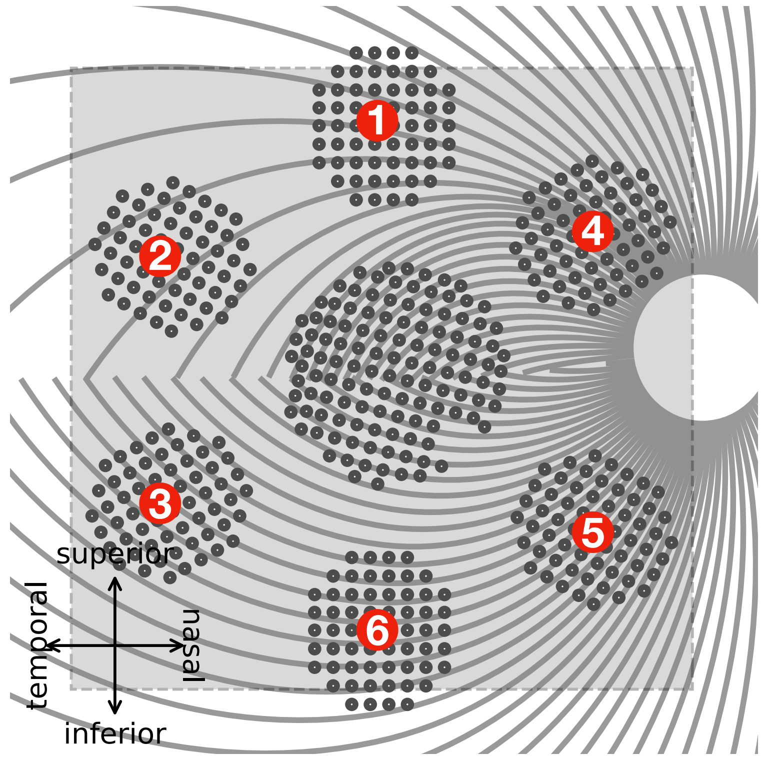
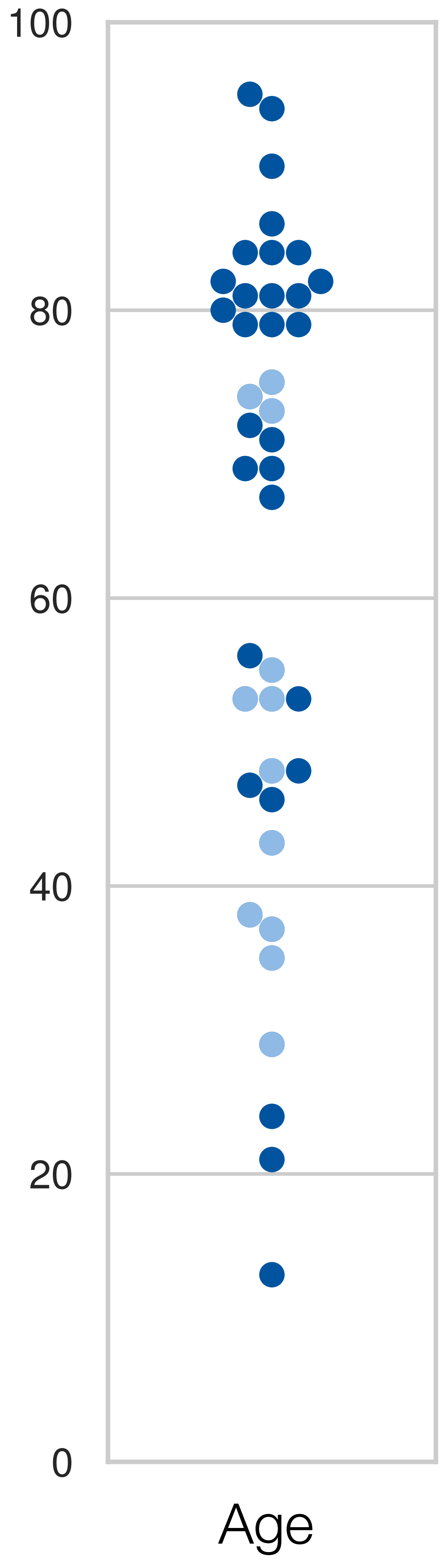




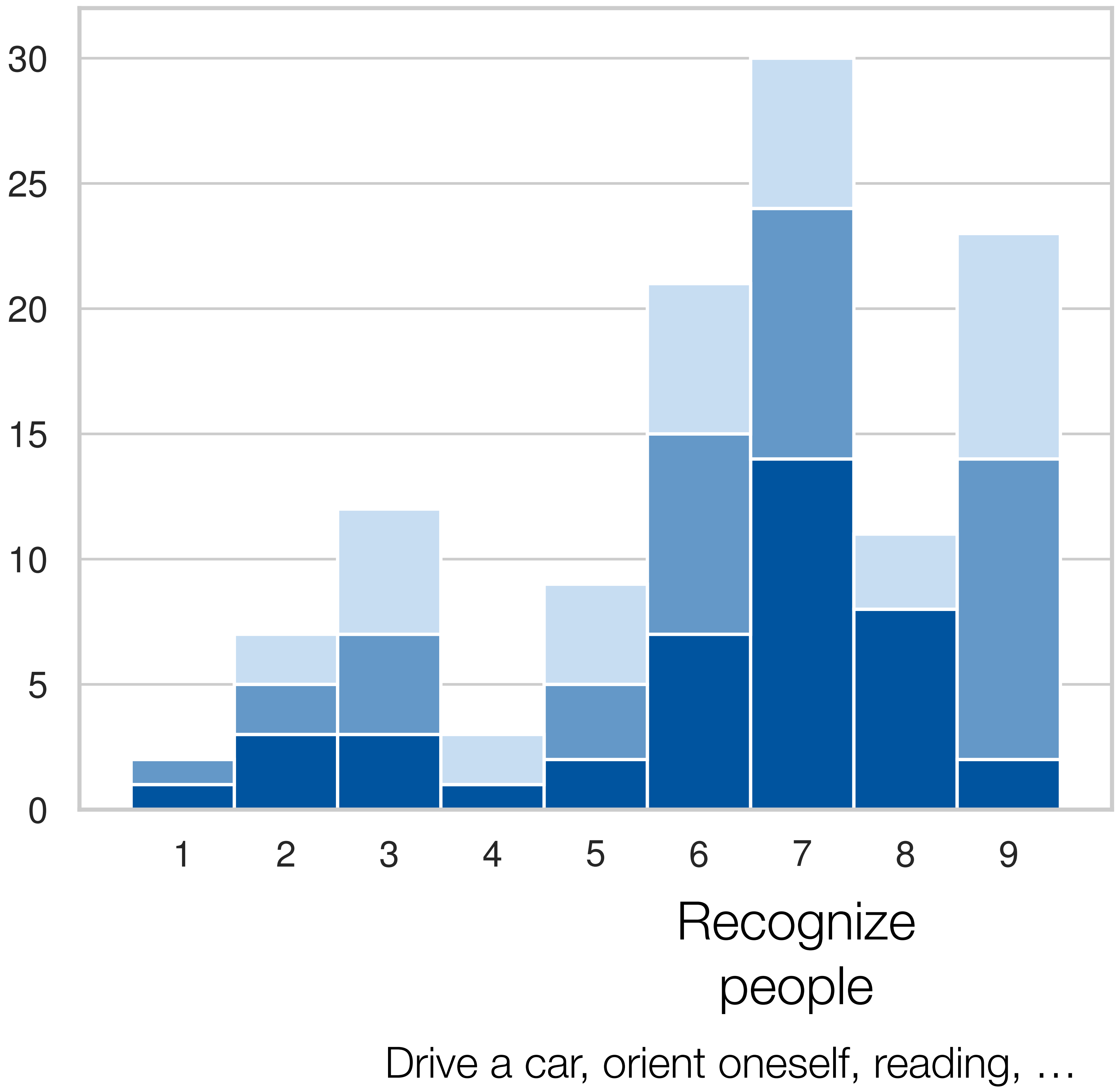




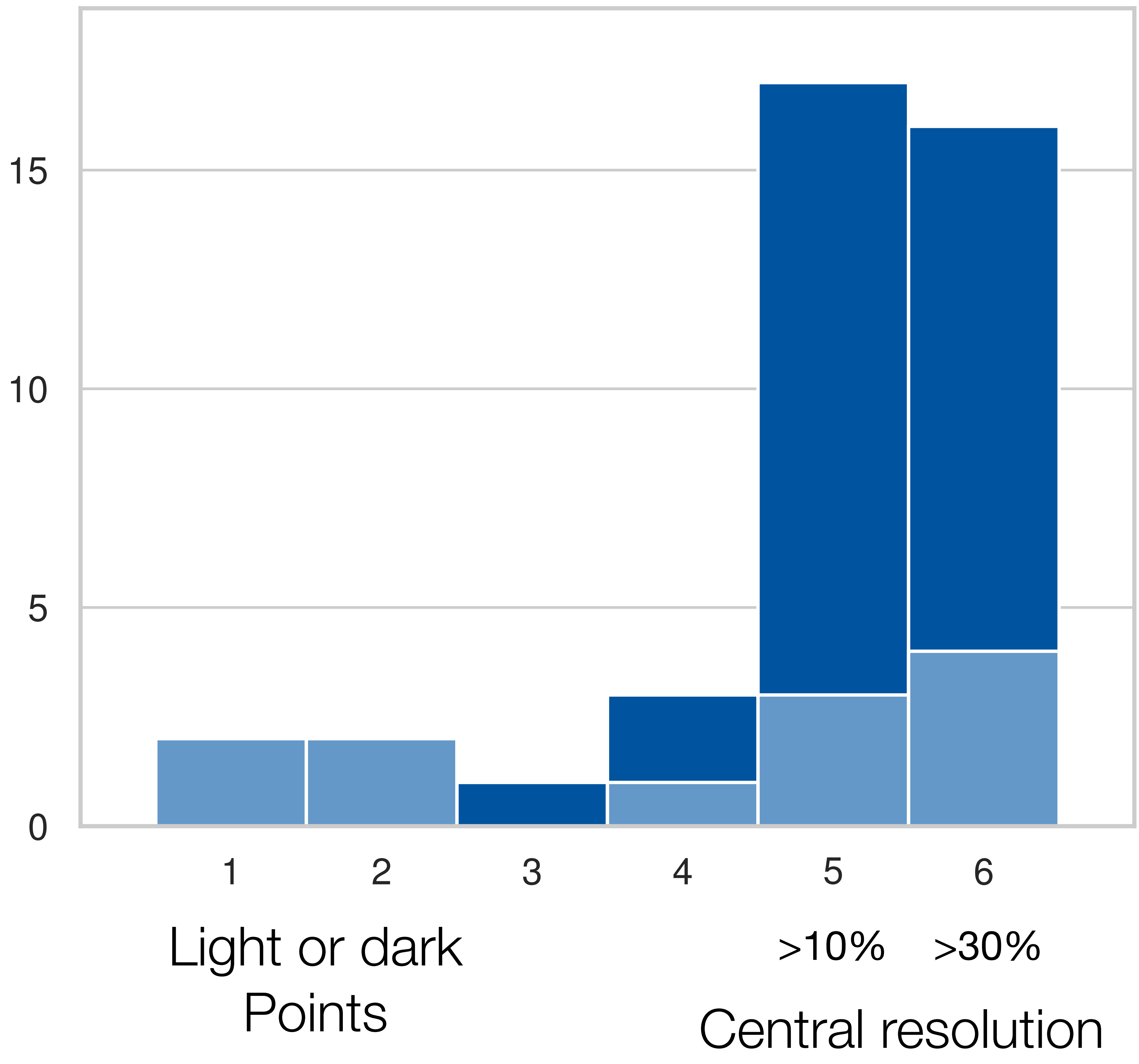
Survey of Individuals with Visual Impairment



Preferred improvements in general



Essential functionalities for a treatment



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